Failure Metabolism

Why persistent machine intelligence should digest failure into structured memory instead of merely logging it

Abstract

Modern AI and industrial software stacks are rich in telemetry yet surprisingly poor at converting failure into durable operational knowledge. Logs describe events, traces connect execution spans, and agent memories preserve selected context, but these mechanisms often stop short of turning a failed action into reusable evidence about what the system should believe or do differently next time. This paper proposes **failure metabolism**: an architectural pattern in which failure is captured as a first-class, causally linked object, evaluated against subsequent recovery and outcome evidence, and assimilated into persistent system memory. The goal is not autonomous self-modification. The goal is a system that becomes progressively harder to fool in the same way twice.

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Author	Mark Yarian
Affiliation	Yarian Works, LLC / SynapticSteel
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Thesis: an intelligent system should not merely survive failure; it should preserve the causal evidence necessary to reinterpret future situations.

This draft intentionally separates observed industry practice, cited research, and proposed SynapticSteel architecture. It is designed for editing before public release.

1. The problem: systems remember events, not meaning

Industrial systems already collect enormous quantities of evidence: operating-system events, application logs, metrics, traces, alarms, maintenance records, model outputs, network observations, and human decisions. Yet a record of what happened is not automatically a memory of what was learned. A timestamped exception can prove that a component failed; it does not, by itself, preserve why the failure mattered, which hypothesis was rejected, what recovery worked, or whether the same evidence should alter a future decision.

This gap is becoming more consequential as AI agents gain access to tools and physical or enterprise systems. Recent work on execution provenance argues that final-answer accuracy is insufficient for understanding agent behavior because evidence, memory, tool calls, environment observations, intermediate claims, and actions must be connected across an execution trajectory [1]. Likewise, current manufacturing research is exploring memory-enhanced multi-agent systems in which monitoring, decision support, and memory governance are coordinated rather than treated as unrelated functions [2].

A useful distinction

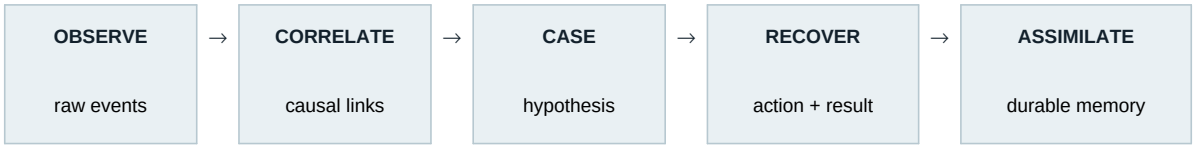
Mechanism	Primary question	What is still missing
Log	What happened?	Meaning, causal confidence, recovery effectiveness
Trace	How did execution flow?	Long-horizon operational learning
Alert	What needs attention?	Durable evidence after the alert closes
Agent memory	What context should be recalled?	Authoritative lineage and mutation history
Failure metabolism	What should this failure change?	Proposed synthesis

The proposed shift is subtle: failure stops being an endpoint in observability and becomes an input into the system's future interpretation of evidence.

2. Failure metabolism as an architectural pattern

The biological metaphor is useful only if it produces engineering constraints. Metabolism does not mean that the system 'feels' failure. It means that raw events are transformed into reusable internal material. A failure-metabolism pipeline therefore needs a deterministic path from observation to evidence, from evidence to interpretation, from interpretation to disposition, and from disposition to future retrieval.

Proposed processing loop



Observe. Capture the original event without rewriting it into a convenient narrative. **Correlate.** Connect it to the software version, model, node, sensor, operator, configuration, and preceding events that were actually present. **Case.** Form a bounded investigation object containing hypotheses and evidence references. **Recover.** Record what intervention was attempted and what happened afterward. **Assimilate.** Promote only validated lessons into durable operational memory.

The final step is the one ordinary logging architectures usually omit. Assimilation should not mean editing history. Original evidence remains immutable; the system adds a later interpretation linked to that evidence. A disproven interpretation should itself remain visible. In this model, correction is additive rather than revisionist.

Minimal failure object

Field	Purpose
failure_id / case_id	Stable identity across services and time
evidence_refs	Pointers to immutable or integrity-checked source evidence
causal_parent	What event or decision directly preceded this object
hypotheses	Competing explanations with confidence and status
intervention	Human or machine action actually attempted
outcome	Observed post-intervention result
disposition	unresolved / mitigated / disproven / assimilated
future_retrieval_keys	Conditions under which this lesson should be recalled

3. Why event sourcing and provenance matter

Failure metabolism depends on an uncomfortable rule: the system must preserve enough history to reconstruct how it reached a conclusion. That pushes the design toward event-sourced and provenance-oriented techniques rather than mutable 'current state' alone.

This direction is not hypothetical. Recent agent research proposes event sourcing specifically to separate probabilistic cognitive intention from deterministic state mutation, using structured intentions, append-only events, materialized views, and replay verification [3]. Other work argues for evidence tracing and execution provenance as the bridge from agent traces to auditability and trust [1]. An IEEE industrial paper published this year similarly treats lifecycle provenance as a traceability problem spanning datasets, models, metrics, validation, and human approvals [4].

Design consequence: memory should be layered

Layer	Character	Example
Evidence	Append-only or integrity-checked	raw event, packet fact, sensor sample, tool output
Projection	Rebuildable	current health, topology, active case state
Interpretation	Versioned and challengeable	diagnosis, confidence, causal hypothesis
Disposition	Governed	approved change, rejected hypothesis, human decision
Learning	Derived from outcomes	reliability score, retrieval rule, recurring signature

This layering prevents a reasoning model from becoming the authority merely because it produced the latest explanation. Models propose interpretations; durable evidence and governed disposition determine what the system is allowed to treat as operational truth.

4. An industrial example

Consider a vibration-sensitive production machine. A sensor pod reports an increase in a narrow vibration band. A maintenance model proposes bearing wear. Minutes later, network telemetry shows intermittent controller latency. An acoustic classifier finds no corresponding bearing signature. A human technician reseats a loose mounting bracket, after which the vibration anomaly disappears.

A conventional system may close the alert as 'resolved.' A failure-metabolizing system retains a more useful structure: the original vibration observation; the model's bearing hypothesis; contradictory acoustic evidence; the network condition as a potentially unrelated concurrent anomaly; the human intervention; and the measured post-intervention return to baseline.

What the system can learn without pretending certainty

Candidate lesson	Status after one case
This spectral pattern proves bearing wear.	Disproven in this case
Mounting looseness can imitate this vibration signature.	Supported, low-to-moderate confidence
Acoustic disagreement is useful when evaluating bearing hypotheses.	Supported as a retrieval cue
Controller latency caused the vibration anomaly.	Unsupported; preserve as concurrent evidence only
A similar future case should retrieve this incident.	Reasonable operational memory rule

> **The objective is not to transform one anecdote into a universal rule. It is to make the anecdote retrievable as structured counterevidence the next time a similar hypothesis appears.**

Over many cases, the system can begin estimating which diagnostic sources are reliable under which conditions. That is a much stronger form of operational learning than simply counting alerts or storing chat summaries.

5. Relationship to SynapticSteel

SynapticSteel is relevant here because several existing design choices already point toward this pattern, but the complete failure-metabolism loop should be treated as a proposed unification rather than claimed as a finished capability.

Existing or previously established design direction

The architecture uses durable stores and APIs for persistent truth while ZeroMQ is reserved for transient coordination. Code Genome is intended to preserve software meaning and lineage. Diagnostics produce evidence about system behavior. Package-based change control keeps machine-initiated changes behind human authorization rather than allowing reasoning components to mutate infrastructure directly. These are compatible foundations for failure metabolism because they separate evidence, signaling, interpretation, and action authority.

Proposed next architectural step

The missing unification is a cross-subsystem case object that can bind an observation to software lineage, diagnostics, hypotheses, interventions, package outcomes, human dispositions, and later recurrence. That object should be model-independent. A local language model, temporal network, deterministic diagnostic, sensor classifier, or human operator should all be able to contribute evidence without becoming the owner of the case.

SynapticSteel subsystem	Potential role in failure metabolism
Code Genome	Software provenance and meaning lineage
Diagnostics	Evidence generation and validation history
Phase Atlas	Intent, planned work, blockers, and lifecycle context
Package Manager	Governed action and outcome capture
Cartographer	Network/topological observations
Sensor pods	Physical-world observations
ZeroMQ fabric	Transient 'evidence ready / work ready' signaling
Persistent APIs / ledgers	Canonical durable record

6. Risks, counterarguments, and limits

Memory can fossilize mistakes. A durable system can become confidently wrong if weak interpretations are promoted too early. The architecture therefore needs explicit confidence, challenge, supersession, and disconfirmation semantics.

Event stores can become enormous. Keeping everything forever is not intelligence. Raw high-volume telemetry may need retention tiers, compression, summaries, and content-addressed evidence bundles while preserving enough source material to audit important conclusions.

Causality is easy to overclaim. Temporal adjacency is not causation. The system should distinguish recorded causal links, inferred relationships, correlations, and unresolved coincidence.

Human approval is not automatically correct. Human decisions should be recorded as authoritative dispositions where policy requires them, but they should remain distinguishable from empirical evidence.

LLMs should not rewrite the ledger. Language models are useful for hypothesis generation, synthesis, retrieval, and critique. They are poorly suited to silently deciding which inconvenient evidence should disappear.

Evaluation questions for a prototype

Question	Possible metric
Can a past failure be reconstructed?	percentage of cases with complete evidence lineage
Does prior failure improve future diagnosis?	change in time-to-correct-hypothesis / false diagnosis rate
Can incorrect lessons be reversed?	supersession and disconfirmation success
Does the system preserve uncertainty?	rate of inferred edges incorrectly presented as facts
Can a model be replaced without losing memory?	case continuity across model substitution

7. Conclusion

AI systems are becoming increasingly capable of acting, but action without durable causal memory creates a recurring engineering failure mode: the system can make the same category of mistake repeatedly while producing excellent logs about each occurrence.

Failure metabolism reframes operational memory around a stricter question: **what should this event change about future interpretation?** The proposed architecture preserves raw evidence, connects it to causal and contextual lineage, records competing hypotheses, measures interventions against outcomes, and only then promotes validated lessons into future retrieval.

> **For persistent distributed intelligence, this property may be more important than giving agents increasingly elaborate conversations. A system becomes operationally more intelligent when yesterday's failure changes tomorrow's behavior without requiring yesterday's model to still exist.**

References

[1] Wang, Y. et al. (2026). *From Agent Traces to Trust: Evidence Tracing and Execution Provenance in LLM Agents.* arXiv:2606.04990.

[2] Li, D. et al. (2026). *An operations memory-enhanced multi-agent system for human-centric manufacturing process monitoring and decision support.* CIRP Annals, 75(1), 625-629. DOI: 10.1016/j.cirp.2026.04.080.

[3] dos Santos Filho, E. B. (2026). *ESAA: Event Sourcing for Autonomous Agents in LLM-Based Software Engineering.* arXiv:2602.23193.

[4] *Blockchain-Based Provenance for Artificial Intelligence Lifecycle Traceability.* 2026 IEEE International Conference on Industrial Technology. DOI: 10.1109/ICIT64854.2026.11490281.

Publication note

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SynapticSteel is a Yarian Works, LLC initiative exploring persistent, distributed intelligence across software, infrastructure, and physical systems.